

Assignment Sheet

Task 1 – Dependencies

(8+4+4+4 Points)

a) Fill the table to show the dependencies in the following code:

```
I1: add t0, t1, t2
I2: mul t2, t0, t1
I3: mulhsu t3, t0, t1
I4: seqz t4, t3
I5: xori t4, t4, -1
I6: add t4, t4, t0
```

	I1	I2	I3	I4	I5	I6
I1	-					
I2	-	-				
I3	-	-	-			
I4	-	-	-	-		
I5	-	-	-	-	-	
I6	-	-	-	-	-	-

- b) Decide if the following re-ordered sequence can be executed without interfering with any dependencies. If not, which dependencies invalidate the sequence?

I1, I4, I2, I3, I5, I6

Your answer:

- c) Execute the code in a 5-stage pipeline without forwarding and branch-prediction. How many NOPs are needed?

Your answer:

- d) How many NOPs are needed in the same pipeline with forwarding from the WB- to the ID-stage?

Your answer:

This assignment is due on 12.12.2025 at 10 am.